LUMAFLOW · DESIGN DECISION RECORD

LumaFlow recommendations accepted

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DOCUMENT CONTROL

Document ID	LMF-DDR-002	Version	0.3
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/lumaflow		
Source	docs/decisions/0002-recommendations-accepted.md @ 8a57af4		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	O1 decided by Amish as recommended (LMF-DEC-001)
0.2	2026-09-26	Amish Chadha	Budget top-up approved by Amish; O4 decided ($340)
0.1	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); record the decided TRL 3 items, the design changes they caused and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted for items E1 to E8; O4 decided on 2026-09-26 (budget top-up to $340); item O1 decided by Amish on 2026-10-02: "i approve your recommendations for all 555 open decisions." (LMF-DEC-001)

Context

After the TRL 3 session, `docs/REVIEW.md` (session 2026-09-25, TRL 3) and LMF-DDR-001 (O2, O3) listed eight items as "Proposed, awaiting Amish". Seven carried a recommendation, and the partner question (O1) did not. On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Every item with a recommendation is therefore decided in favor of it; where the recommendation named several options, the recommended option is the decision. Items without a recommendation stay open. TRL 4 remains on hold by Amish's instruction, so anything that needs a build, a measurement, firmware beyond a sketch or purchasing is recorded as decided but on hold.

Options considered

The options for each item are those listed in `docs/REVIEW.md` (TRL 3 session) and in LMF-CAL-001 v0.1, section J.

Decision

Table 1. Decided items. Each is "Decided by Amish, 2026-09-25: go with recommendation".

#	Item	Decision	What changed in the repo
E1	End cap material (DDR-001 O2)	Confirm a 316 stainless lower cap and an acetal upper cap shielded by the PTFE liner, top disc and a 316 outlet insert	Wording only; LMF-PRC-001 v0.4 marks the choice as decided. The design already used these materials
E2	Route to close R2	Option (d), a wider bore of about 50 mm, combined with option (c), a high-reflectance liner, plus a modest flow cut; lower flow alone (a) stays the fallback	`cad/src/model.py` : bore 25 to 50 mm, liner 34 to 59 mm OD, tube 40 to 65 mm OD, caps 60 to 90 mm, window 32 x 6 to 57 x 10 mm on a 51 mm seat, LED circle 16 to 32 mm, sink 64 to 90 mm, M4 to M5 tie rods. Liner reflectance 0.80 to 0.95. Design flow 2.0 to 1.2 L/min, the largest round flow that keeps the 20 % margin (limit 1.30 L/min, LMF-CAL-001 v0.2, J1). RED at 90 %/cm 14.7 to 19.0 became 51.9 to 76.3 mJ/cm²; R2 not met became met. R1 restated to 1.2 L/min
E3	Dose monitor position (R4)	Move the photodiode from the top disc to the tube wall at mid height and keep the proportional mapping	Photodiode, sensor window and a clamp-on saddle at Z = 166 mm in the model; top disc now solid. Signal at 90 %/cm 2.4 to 16.3 nA; at 70 %/cm unresolved to 0.38 nA. R4 stays at risk (firmware and test needed). The optional second reference sensor was not added
E4	Budget (R16)	Option (c): hold the budget until the R2 route is chosen, then re-price	BOM re-priced for the 50 mm reactor: $249.00 to $338.00. The recommendation named no figure that covers this total (option (a) was $250), so `budget_usd` stays at $225 and the new figure is an open item (O4)
E5	Flow sensor pulse rate (R5)	A sensor of 15 Hz per L/min or more	BOM line 8 and the calc: worst-case switch-on 0.46 to 0.24 s; R5 at risk became met
E6	R8 wording	State that the 0.5 bar excludes the flow restrictor; confirm the sensor and valve flow coefficients from datasheets	R8 restated in LMF-REQ-001 v0.4. Pressure drop 0.42 to 0.15 bar at the new flow; R8 at risk became met. Datasheet values are to be read when specific parts are chosen
E7	Pressure limiter	An upstream pressure-limiting valve is an installation requirement	New R17 (limiter at 4 bar or less). Window 3.09 MPa at 4 bar and 6.18 MPa at a transient of twice the setting, against 6.8 MPa. Installation part, not in the BOM, like the sediment pre-filter
E8	Thermal cut-back	A board temperature sensor that cuts LED current above 50 °C	NTC on the LED board (BOM line 4) and a dimming input on the driver (line 10); R10 restated. Board 44.1 to 55.9 °C became 37.1 to 44.8 °C with the larger sink and cap; R10 at risk became met. The cut-back is a firmware rule, documented only

Table 2. Items left open on 2026-09-25, since decided.

#	Item	Why it stays open
O1	Partner that can supply real well-water UVT data (DDR-001 O1)	No recommendation was made on 2026-09-25. Decided by Amish, 2026-10-02, as later recommended: a university extension programme for private well owners, with the Texas A&M AgriLife Extension Texas Well Owner Network as the first candidate to approach (LMF-DEC-001)
O4	Budget figure for the $338.00 BOM	New after re-pricing. Options: (a) raise `budget_usd` to about $340; (b) keep $225 and cut cost, for example a plain PTFE liner (46.3 mJ/cm² at 1.2 L/min, below the 20 % margin) or fewer machined parts; (c) keep $225 and leave R16 visibly not met. **Budget top-up to $340: decided by Amish, 2026-09-26** ("I am ok with the budget top ups"). `budget_usd` is now 340 and R16 is met ($338.00, $2.00 under; LMF-REQ-001 v0.5, LMF-CAL-001 v0.3)

Table 3. Decided but on hold (TRL 4, on hold by Amish's instruction).

Item	Work on hold
E2	Measuring the wetted liner reflectance at 275 nm and biodosimetry of the reactor
E3, E8	Firmware for the dose mapping, the 1 s fail-safe and the thermal cut-back, beyond a labeled sketch
E4	Quotes and purchasing for the re-priced BOM

Consequences

- LMF-PRB-001 v0.4, LMF-PRC-001 v0.4, LMF-REQ-001 v0.4, LMF-CAL-001 v0.2 and LMF-DDR-001 v0.2 record these decisions.
- `cad/src/model.py`, the STEP and STL exports, drawing LMF-DWG-001 (Rev P1 to P2), the concept sheet LMF-DWG-010 (Rev P2), all media and `bom/bom.csv` follow the new design.
- Requirement status (LMF-CAL-001 v0.2): 13 met, 1 at risk (R4), 2 not met (R3, accepted by D1; R16), 1 not verifiable at TRL 3 (R15). Before: 8 met, 4 at risk, 3 not met, 1 not verifiable, of 16.
- `project.yaml` keeps `trl: 3` and `trl_target: 3`; `budget_usd` stayed at 225 until the budget top-up to $340 (O4, decided by Amish on 2026-09-26), after which R16 is met and the count is 14 met, 1 at risk, 1 not met (R3), 1 not verifiable. The pitch is unchanged; it does not name a flow rate.
- No other repo is affected: LumaFlow uses no SwapCell pack or shared part.